Jiarui Liang

ljr23@mails.tsinghua.edu.cn · https://liang-jiarui.github.io/

Research interests: Earth system modeling, atmospheric chemistry, greenhouse-gas budgets, and wildfire emissions

Education

2023.09 – Present **Tsinghua University, Ph.D. Candidate, Shenzhen International Graduate School, Environmental Science and New Energy Technology. Supervisor: Assoc. Prof. Bo Zheng. Research field: Atmospheric Science**

2019.09 – 2023.06 **China University of Petroleum (Beijing), B.Eng., College of Chemical Engineering and Environment, Environmental Engineering**

Research Experience

2025.08 – Present **Evolution of wildfire emissions and associated Earth system impacts**
Ph.D. dissertation
– Using the Community Earth System Model (CESM), quantify the atmospheric impacts (O_3 and $PM_{2.5}$) and climate effects of global wildfire emissions

2024.12 – 2025.08 **Source apportionment of summer ozone in Beijing with a machine learning-enhanced chemical transport model**
Ph.D. research
– Built a machine learning-enhanced chemical transport modeling framework (GEOS-Chem and WRF-CAMx) and quantified summer ozone sources in Beijing with OSAT

2023.09 – Present **Anthropogenic nitrous oxide emissions in China: trends, variations, and drivers over the past 40 years**
Ph.D. research
– Quantify China's anthropogenic N_2O emissions and mitigation potential with an emission inventory and machine learning, and analyze pathways consistent with the 1.5°C target

2022.07 – 2025.02 **Source-sink budgets and spatiotemporal patterns of CO_2, CH_4, and N_2O in China**
Ph.D. research
– Ensemble analysis of China's greenhouse-gas source-sink budgets and spatiotemporal patterns, using global atmospheric inversions from the Global Carbon Project

2021.09 – 2023.09 **Operation optimization of industrial circulating-water systems with variable-frequency drives and fouling**
Undergraduate research
– Built a mixed-integer nonlinear model of the circulating-water system, optimized variable-frequency drive frequencies and valve openings, and solved it with a genetic algorithm

Publications

2026

1. Zhang, D., Zheng, B*., Zhu, B., Xu, Y., Liu, X., Wang, N., Zhang, L., **Liang, J.**, et al. OSCAR-fire v1.0: A fire bookkeeping module for the OSCAR compact Earth system model, *EGUsphere*. https://doi.org/10.5194/egusphere-2026-5536

2024

1. **Liang, J.**, Li, Y., Wang, Y*., et al. Long-term operation optimization of circulating cooling water systems under fouling conditions, *Chinese Journal of Chemical Engineering*. https://doi.org/10.1016/j.cjche.2023.08.008

2023

1. Gu, C., Liu, S., **Liang, J.**, et al. Degradation of 1,2,3-trichloropropane by pyrite activating sodium percarbonate and the implications for groundwater remediation, *Journal of Environmental Chemical Engineering*. https://doi.org/10.1016/j.jece.2022.109217

2022

1. **Liang, J.**, Li, L., Wang, Y*., et al. Operation optimization of existing industrial circulating water system considering variable frequency drive, *Chemical Engineering Research and Design*. https://doi.org/10.1016/j.cherd.2022.08.010
2. **Liang, J.**, Sun, Y., Li, Y., et al. Operation optimization of circulating-water systems with variable-frequency drives under fouling conditions, Process Systems Engineering (PSE) Annual Meeting 2022.